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Rigorous analytical approximation of tritronquée solution to Painlevé-I and the first singularity

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Abstract

We use a recently developed method [1,2] to determine approximate expression for tritronquée solution for P-1: $y'' + 6y^2 - x = 0$ in a domain D with rigorous bounds. In particular we rigorously confirm the location of the closest singularity from the origin to be at $x = -\frac{770766}{323285} = -2.3841687675\cdots$ to within 5×10^{-6} accuracy, in agreement with previous numerical calculations [6].
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1. Introduction

Painlevé equations or transcendents, as they are sometimes referred to, arise quite often in many areas of mathematics; so much so that they have sometimes been referred to as nonlinear special functions. While some properties of solutions to Painlevé equations are known analytically, there are other properties for which only numerical evidence exists. Numerical calculations of these solutions still remain an active area of interest (see for instance [3–5]). However, as far as we are aware, numerical computations thus far have not included rigorous error analysis. Even for the P-1 equation, for the special solutions referred to as Tritronquée solutions [8,9], the location of singularity closest to the origin, while computed with apparent high precision [6], has not been justified rigorously.

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The purpose of this paper is (i) to determine rigorous error bounds for the location of the first singularity for tritronquée solution to P-1, and (ii) to demonstrate more generally how a method developed earlier in the context of proof [2] of the Dubrovin Conjecture for P-1 can be extended to obtain rigorous error bounds on approximate analytical expression for solution in different parts of the complex plane. Note however that though the method used here is similar to [2], which is related to the method originally developed in [1], prior analysis did not concern accurate calculation of solution in any part of the domain. In describing the analysis, it will also be apparent that the method generalizes to other solutions of P-1 and to other differential equations.

2. Problem

The tritronquée solution[1] to the Painlevé-1 equation:

$$\frac{d^2y(x)}{dx^2} + 6y^2(x) - x = 0, \tag{1}$$

on $\mathbb{C}$ is the unique solution with the asymptotic behavior

$$y = \sqrt{\frac{x}{6}}\left[1 + o\left(x^{-5/8}\right)\right] \text{ as } x \to +\infty. \tag{2}$$

It is well-known that any solution to P-1 is single valued and meromorphic in $\mathbb{C}$, where singularity locations (in the form of a double poles) depend on initial conditions on (y, y'). Instead of initial conditions, the solution is also completely characterized by the location of a singularity x_p and the coefficient of $\hat{a}_2$ and has the locally convergent series representation:

$$y(x) = -\frac{1}{(x-x_p)^2} + (x-x_p)^2 \sum_{j=0}^{\infty} \hat{a}_j (x-x_p)^j \tag{3}$$

where $\hat{a}_0 = -\frac{x_p}{10}$, $\hat{a}_1 = -\frac{1}{6}$, $\hat{a}_3 = 0$ and for $n \geq 4$, $\hat{a}_n$ is determined from the recurrence relation

$$\hat{a}_n = -\frac{6}{(n+5)(n-2)} \sum_{j=0}^{n-4} \hat{a}_k \hat{a}_{n-4-k} \tag{4}$$

The location of x_p closest to the origin for the tritronquée solution is known [6] to be on the negative real axis, and numerical calculations [6] suggest its location, though this has not been confirmed rigorously. Singularities at large distance from the origin can be rigorously estimated based on adiabatic invariance of conserved quantities [10]. When x_p is not particularly large, like the first singularity of the tritronquée solution, we are unaware of any method of rigorous analysis to confirm its location.

[1] There are actually five different Tritronquee solutions, each corresponding to different choice of anti-Stokes line where one demands $y - \sqrt{\frac{x}{6}} = o(x^{-1/8})$ however, they are all related to each other through rotation of dependent and independent variables.

The main result of this paper is Theorem 2.3 given below. However, we first define a few quantities.

Definition 2.1. We define $r = \frac{7}{10}$, $x_0 = -\frac{770766}{323285} = -2.3841687675\cdots$, $L = \frac{11}{2}$, $L_0 = -\frac{49}{100}$, $b = \frac{4}{5}(24)^{1/4}$ and $a = \frac{5}{2}b$,

$$\tau = \tau(x) := \frac{x - (L + x_0 + r)/2}{(L - x_0 - r)/2} \tag{5}$$

and P_u and P each to be polynomials:

$$P_u(\tau) = \sum_{k=0}^{22} c_k \tau^k \,, \tag{6}$$

where $\mathbf{c} := (c_0, c_1, \cdots, c_{22})$ is given by

$$\begin{aligned}\mathbf{c} = \Bigg(&\frac{335867}{539062}, \frac{419712}{989125}, -\frac{352463}{3539236}, \frac{60789}{1703279}, -\frac{132842}{11825541}, \frac{43961}{54574472}, \frac{39599}{12036926},\\ &-\frac{213665}{48625258}, \frac{61644}{14973337}, -\frac{107283}{33444500}, \frac{44761}{18892011}, -\frac{28249}{13550715}, \frac{20641}{14839893}, \frac{13459}{92774551},\\ &-\frac{4992}{34838093}, -\frac{11771}{8149937}, \frac{24115}{27631671}, \frac{42106}{39550107}, -\frac{21163}{32637441}, -\frac{9782}{15918509}, \frac{11581}{32652169},\\ &\frac{14692}{88640147}, -\frac{12278}{123249611}\Bigg)\end{aligned} \tag{7}$$

$$P(\zeta) = \sum_{k=0}^{17} a_n \zeta^n \,, \tag{8}$$

where $a_0 = -x_0/10$, $a_1 = -1/6$, $a_2 = \frac{19949}{321055}$, $a_3 = 0$ and

$$a_n = -\frac{6}{(n+5)(n-2)} \sum_{j=0}^{n-4} a_k a_{n-4-k} \quad \text{for } 17 \geq n \geq 4 \tag{9}$$

Further, we define

$$N_0(x) = -\frac{4412401}{98304\sqrt{6}} x^{-19/2} \left[1 - \frac{1225}{90049\sqrt{6}} x^{-5/2} + \frac{30625}{2161176} x^{-5}\right] \tag{10}$$

$$\mathcal{G}_1(x) = x^{-5/8} \exp\left[-ibx^{5/4}\right] , \mathcal{G}_2(x) = x^{-5/8} \exp\left[ibx^{5/4}\right] , \tag{11}$$

$$w_0 = \sum_{j=1}^{2} \frac{(-1)^j}{ia} \mathcal{G}_j(x) \int_{\infty}^{x} \mathcal{G}_{3-j}(y) y N_0(y) dy. = \Re\left\{\int_0^{\infty} e^{-sbx^{5/4}} \mathcal{W}_0\left(x^{5/4}, s\right) ds\right\} , \tag{12}$$

where

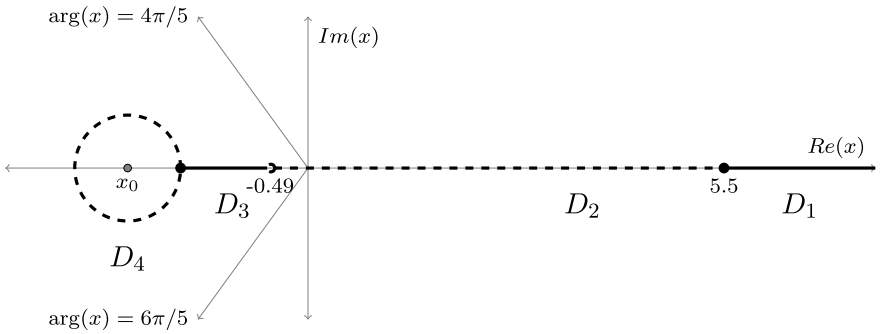


Fig. 1. Sketch of domain $D = D_1 \cup D_2 \cup D_3 \cup D_4$.

$$\mathcal{W}_0(z,s) = -\frac{4412401\sqrt{6}}{368640az^7}\left((1+is)^{-15/2} - \frac{1225\sqrt{6}}{540294z^2}\,(1+is)^{-19/2} + \frac{30625}{2161176z^4}\,(1+is)^{-23/2}\right) \tag{13}$$

where in (13) the principal branch is used, i.e. $(1+is)^{-\kappa} = 1$ at $s = 0$.

Remark 2.2. Integration by parts of the Laplace transform representation of w_0 and its derivative in (12) relates to error functions with complex arguments, which are considered known in the sense it may be calculated to any desired precision [11]. For instance $w_0(L) = -1.17414\cdots \times 10^{-7}$, $w_0'(L) = 2.03367\cdots \times 10^{-7}$.

We also define domains D_j for $j = 1, \cdots, 4$ with $D_1 = [L, \infty)$, $D_2 = [L_0, L)$, $D_3 = [x_0 + r, L_0)$ and $D_4 = \{x \in \mathbb{C} : |x - x_0| = r, x \neq x_0 + r\}$. We define $D = D_1 \cup D_2 \cup D_3 \cup D_4$ (see Fig. 1).

Theorem 2.3. *Let*

$$y_0(x) = \begin{cases} \sqrt{\frac{x}{6}}\left[1 + \frac{1}{8\sqrt{6}}x^{-5/2} - \frac{49}{768}x^{-5} + \frac{1225}{1536\sqrt{6}}x^{-15/2} + w_0(x)\right] & \text{on } D_1 \\ -\frac{1}{(x-x_0)^2} + P_u(\tau(x)) & \text{on } D_2 \cup D_3 \\ -\frac{1}{(x-x_0)^2} + (x-x_0)^2 P(x-x_0) & \text{on } D_4 \end{cases} \tag{14}$$

Then the tritronquée solution y has the representation

$$y(x) = y_0(x) + E(x)\ , \text{where } \left|E(x)\right| \le 2.35 \times 10^{-5}\ , \left|E'(x)\right| \le 1.16 \times 10^{-4} \tag{15}$$

Moreover, y has a unique double pole singularity at $x = x_p \in \{\zeta : \zeta \in \mathbb{C}, |\zeta - x_0| < r\}$ with $|x_p - x_0| \le 4.1 \times 10^{-6}$. This is the closest singularity of the tritronquée solution from the origin.

Note: The proof of the theorem is completed in Section 7 after establishing some preliminary results.

Remark 2.4. From evaluation of expression for y_0, y_0' at the points of discontinuity, it is readily checked that $\left|y_0(L^+) - y_0(L^-)\right| \le 5 \times 10^{-14}$, $\left|y_0'(L^+) - y_0'(L^-)\right| \le 7.5 \times 10^{-14}$, $\left|y_0\left([x_0+r]^+\right) - y_0\left(x_0 + re^{i0^+}\right)\right| \le 4 \times 10^{-10}$, $\left|y_0'\left([x_0+r]^+\right) - y_0'\left(x_0 + re^{i0^+}\right)\right| \le 7 \times 10^{-8}$.

Remark 2.5. The strategy we pursue is as follows: For each D_j ($j = 1, 2, 3, 4$), a choice[2] of y_0 is made to ensure that the remainder $R(x) := y_0''(x) + 6y_0^2(x) - x$ is small and the mismatches $y_0(x_e^-) - y_0(x_e^+)$, $y_0'(x_e^+) - y_0'(x_e^-)$ when boundary point x_e is approached from domains D_j and D_{j+1} is either zero or very small (see Remark 2.4). Then the differential equation satisfied by $E = y - y_0$, with initial condition at the end point, is transformed to an equivalent nonlinear integral equation in the form $E = \mathcal{N}[E]$. A convenient Banach space $\mathcal{S} \subset C^0(D_j)$ is defined and a small enough ball $\mathcal{B} \subset \mathcal{S}$ centered at $E_0 = \mathcal{N}[0]$ for which $\mathcal{N}$ is contractive. The Banach contraction mapping theorem ensures existence and uniqueness of such $E \in \mathcal{B}$ satisfying the integral equation, and this result is used to estimate bounds on $|E|$ and $|E'|$. In particular, this allows for error analysis in domain D_{j+1}, once error estimates are completed in D_j since initial conditions in D_{j+1} are then determined to within small errors when continuity of $y = y_0 + E$ and its derivative is demanded at the end point x_e.

$$E(x_e^-) = E(x_e^+) + y_0(x_e^+) - y_0(x_e^-) \quad , \quad E'(x_e^-) = E'(x_e^+) + y_0'(x_e^+) - y_0'(x_e^-)\,. \tag{16}$$

For each j, $C^2(D_j)$ regularity of E (and therefore of y) follows using the smoothness of the Kernel in the integral reformulation, and therefore E satisfies the differential equation. This implies that $y = y_0 + E$ satisfies P-1 in each domain D_j and by continuity of y and y' at common end points x_e, it is the same solution to P-1 in D. Since asymptotic condition at ∞ has been enforced, this must be the tritronquée solution.

It is noted that although the proof of the estimates relies on a priori knowledge about local meromorphic representation, it is not essential in the error representation along any particular path in the complex-plane. For instance, if (3) were not available, then on a circle $\zeta - x_0 = re^{iv}$, we could have used a Chebyshev representation in $\cos(v)$ for $\Re(y)$ and $\Im(y)/\sin(v)$ to obtain a representation for solution. In the case the singularity x_0 were not a pole, but say a branch point of order $1/2$, then a Chebyshev representation in $\cos(v/2)$ for $\Re(y)$ would be more appropriate assuming that r is small enough so that we do not run into other singularities on the Riemann surface. The Cauchy integral formula can be extended on Riemann surface to obtain singularity location (for instance see [12]). Other generalizations of the quasi-solution approach to non-integrable systems appear in [13].

Remark on Notation: The framework of the proof in each section is quite similar. Therefore, to avoid proliferation of symbols, we found it convenient to use the same notation for similar quantities in each section. Thus, for each subdomain D_j ($j = 1, 2, 3, 4$); the approximate solution and the error terms are always denoted by y_0 and E respectively.[3] G_1, G_2 will always denote two independent solutions of a homogenous second order differential equation $\mathcal{L}E = 0$ in each

[2] The choice relies on asymptotic series on D_1, Chebyshev polynomial approximation on $D_2 \cup D_3$ and Taylor-like polynomial on D_4.

[3] In §2, it is more convenient to take $E = \sqrt{\frac{x}{6}}(w - w_0)$ and analyze w.

domain, where $\mathcal{L}$ is typically obtained by linearizing P-1 about y_0. Sometimes it is more convenient to choose $\mathcal{L}$ as an operator close to the linearized P-1, but not quite the same. Using G_1, G_2, the differential equation for E with initial condition is transformed to an integral equation in the form $E = \mathcal{N}[E]$. We always denote $E_0 = \mathcal{N}[0]$ and $\mathcal{S}$ is the Banach space of continuous functions in domain D_j equipped with either sup norm or weighted sup-norm, while $\mathcal{B} \subset \mathcal{S}$ is the generic notation for some small ball where $\mathcal{N}$ is contractive. Note that the actual definitions of y_0, E, G_1, G_2, $\mathcal{N}$, E_0, $\mathcal{S}$ and $\mathcal{B}$ differ from section to section.

3. Analysis on $D_1 = [L, \infty)$

It is well-known (for instance see [7, §6.6a]) that if y satisfies (1) and (2), then it is uniquely expressed as[4]:

$$y(x) = \sqrt{\frac{x}{6}}\left[1 + \frac{1}{8\sqrt{6}}x^{-5/2} - \frac{49}{768}x^{-5} + \frac{1225}{1536\sqrt{6}}x^{-15/2} + w(x)\right] \tag{17}$$

where $w(x) = o(x^{-15/2})$. Substituting (17) into (1) and subtracting $[25w(x)]/[64x^2]$ from both sides yield:

$$\mathcal{L}w := w'' + \frac{w'}{x} + \left[2\sqrt{6x} - \frac{25}{64x^2}\right]w = N(w(x), x), \tag{18}$$

where

$$N(w, x) = N_0(x) - \frac{25}{64}x^{-2}\left[1 - \frac{49}{25\sqrt{6}}x^{-5/2} + \frac{49}{12}x^{-5}\right]w - \sqrt{6x}w^2, \tag{19}$$

where $N_0(x)$ is defined in (10). Note that $N(0, x) = N_0(x)$. It is observed that $\mathcal{G}_1$, $\mathcal{G}_2$ defined in (11) are independent solutions to the homogenous equation $\mathcal{L}w = 0$, defined in (18). From the method of variation of parameter, (18) is equivalent to

$$\begin{aligned} w(x) &= \frac{1}{ia}\sum_{j=1}^{2}(-1)^j \mathcal{G}_j(x)\int_{\infty}^{x} t\mathcal{G}_{3-j}(t)N(w(t), t)dt + \alpha_0\mathcal{G}_1(x) + \beta_0\mathcal{G}_2(x), \\ &=: \mathcal{N}[w](x) + \alpha_0\mathcal{G}_1(x) + \beta_0\mathcal{G}_2(x) \end{aligned} \tag{20}$$

up to a convenient choice of α_0 and β_0. The condition $w(x) = o(x^{-15/2})$ is observed to imply $\mathcal{N}[w] = O(x^{-10})$ (see (25) and (27) in the ensuing), implying $\alpha_0 = 0 = \beta_0$ in (20).

Remark 3.1. We look for solution $w \in C(D_1)$ satisfying

$$w(x) = \mathcal{N}[w](x), \tag{21}$$

[4] First four terms on an asymptotic series of y are used. It is observed that including more or less number of terms results in larger error bounds.

with $w(x) = o(x^{-15/2})$ as $x \to \infty$. From regularity of kernel and a standard bootstrap argument $w \in C^2(D_1)$. Then it follows that y in (17) will satisfy (1) and (2).

Definition 3.2. Define Banach space

$$\mathcal{S} := \{f \in C(D_1) : \|f\|_{10} < \infty\} \text{ where } \|f\|_{10} := \sup_{x \in D_1} |f(x)| x^{10}. \tag{22}$$

For $f_0 \in \mathcal{S}$ and $\epsilon > 0$, define

$$\mathcal{B}(f_0, \epsilon) := \{f \in \mathcal{S} : \|f - f_0\|_{10} \le \epsilon\}. \tag{23}$$

We note that with w_0 defined in (12), $\mathcal{N}[0] = w_0$.

Lemma 3.3. $\|w_0\|_{10} \le \frac{15}{2}$ *and* $\left|w_0'(x)\right| \le (8.85)x^{-39/4}$ *on* D_1.

Proof. Using $x^{5/8}\mathcal{G}_j(x) = \exp\left[(-1)^j i b x^{5/4}\right]$, integration by parts in (12) yields:

$$w_0(x) = \frac{8}{5ba}x^{-1/2}N_0(x) + \frac{4}{5ba}\sum_{j=1}^{2}\mathcal{G}_j(x)\int_x^\infty \left[t^{1/8}N_0(t)\right]' t^{5/8}\mathcal{G}_{3-j}(t)dt \tag{24}$$

From explicit computation, it may be checked that for $x \ge L$, $\left|\left[x^{1/8}N_0\right]'\right| = \left[x^{1/8}N_0\right]'$, $|\mathcal{G}_j(x)| \le x^{-5/8}$ for $j = 1, 2$ and $x \ge L$ and therefore it may be checked that $x^{19/2}|N_0(x)|$ is monotonically increasing for $x \ge L$ and using limiting value as $\lim_{x\to\infty}$, we obtain

$$\left|w_0(x)\right| \le \frac{16}{5ab}x^{-5/8}x^{1/8}|N_0(x)| \le \frac{15}{2}x^{-10}. \tag{25}$$

Further, from (24), it follows from using $\mathcal{G}_j(x)\mathcal{G}_{3-j}(x) = x^{-5/4}$ that

$$w_0' = -\frac{1}{ba}x^{-3/2}N_0(x) + \frac{4}{5ba}\sum_{j=1}^{2}\mathcal{G}_j'(x)\int_x^\infty \left[t^{1/8}N_0(t)\right]' t^{5/8}\mathcal{G}_{3-j}(t)dt \tag{26}$$

Note from (11), $|\mathcal{G}_j'(x)| \le \frac{5}{4}bx^{-3/8} + \frac{5}{8}x^{-13/8}$ for $j = 1, 2$. Therefore, again using property $\left|\left[t^{1/8}N_0(t)\right]'\right| = \left[t^{1/8}N_0(t)\right]'$, and $-x^{1/8}N_0(x) = \left|x^{1/8}N_0(x)\right|$ for $t, x \ge L$, it follows from (26) and the fact that $x^{19/2}|N_0(x)| = -x^{19/2}N_0(x)$ is increasing on D_1 and $\lim_{x\to\infty} -x^{19/2}N_0(x) = 18.324\cdots$, we obtain that

$$\begin{aligned}\left|w_0'(x)\right| &\le \frac{1}{ab}x^{-3/2}|N_0(x)| + \frac{8}{5ab}\left[\frac{5}{4}bx^{-3/8} + \frac{5}{8}x^{-13/8}\right]\left|x^{1/8}N_0(x)\right| \\ &\le \frac{2}{ax^{39/4}}|x^{19/2}N_0(x)|\left(1 + \frac{1}{bx^{5/4}}\right) \le \frac{(2)(18.33)}{ax^{39/4}}\left(1 + \frac{1}{bL^{5/4}}\right) \le 8.85x^{-39/4}. \quad \square\end{aligned}$$

Lemma 3.4. *Let $\delta = 2 \times 10^{-3}$. Then there exists a unique fixed point w of $\mathcal{N}$ in $\mathcal{B}(w_0, \delta\|w_0\|_{10})$. Moreover, $\left|w(x) - w_0(x)\right| \le (0.0167)\|w_0\|_{10} x^{-45/4}$ and $\left|w'(x) - w_0'(x)\right| \le 0.29x^{-11}$ on D_1.*

Proof. Let $w \in \mathcal{B}(w_0, \delta\|w_0\|_{10})$. Since $\left[1 - \frac{49}{25\sqrt{6}}x^{-5/2} + \frac{49}{12}x^{-5}\right] \le 1$ on D_1, $|\mathcal{G}_j(x)| \le x^{-5/8}$ for $j = 1, 2$, Lemma 3.3 and $x \ge L$, (21) and (12) imply:

$$\left|\mathcal{N}[w](x) - w_0(x)\right| \le \left[\frac{5(1+\delta)}{68ax^{5/4}} + \frac{16\sqrt{6}}{145ax^{35/4}}(1+\delta)^2\|w_0\|_{10}\right]|\|w_0\|_{10}x^{-10} \le \delta\|w_0\|_{10}x^{-10}. \tag{27}$$

Multiplying by x^{10}, we immediately obtain $\mathcal{N}[\mathcal{B}(w_0, \delta\|w_0\|_{10})] \subset \mathcal{B}(w_0, \delta\|w_0\|_{10})$. Similarly, let w_1 , $w_2 \in \mathcal{B}(w_0, \delta\|w_0\|_{10})$. From the previous arguments, and $\|w_1\|_{10} + \|w_2\|_{10} \le 2(1+\delta)\|w_0\|_{10}$, (12) implies:

$$\begin{aligned}\left|\mathcal{N}[w_1](x) - \mathcal{N}[w_2](x)\right| &\le \left[\frac{5}{68ax^{5/4}} + \frac{16\sqrt{6}}{145ax^{35/4}}2(1+\delta)\|w_0\|_{10}\right]x^{-10}\|w_1 - w_2\|_{10} \\ &\le (0.002)x^{-10}\|w_1 - w_2\|_{10}\end{aligned}$$

Multiplying by x^{10}, shows that $\mathcal{N} : \mathcal{B} \to \mathcal{B}$ is contractive. Thus $w = \mathcal{N}[w]$ has a unique solution $w \in \mathcal{B}(w_0, \delta\|w_0\|_{10})$ as a consequence of Banach fixed point theorem. Since $w = \mathcal{N}[w]$, it follows from estimates in (27) that $\left|w(x) - w_0(x)\right| \le 0.0167\|w_0\|_{10}x^{-45/4}$. On observing $\left[1 - \frac{49}{25\sqrt{6}}t^{-5/2} + \frac{49}{12}t^{-5}\right] \le 1$ on D_1 and using Lemma 3.3, (19) and (10) imply

$$\left|N(w,t) - N_0(t)\right| \le \frac{25}{64t^{12}}\|w_0\|_{10}(1+\delta) + \frac{\sqrt{6}}{t^{39/2}}(1+\delta)^2\|w_0\|_{10}^2 \le \frac{2.94}{t^{12}} + \frac{139}{t^{39/2}}$$

From this result and the bounds $|\mathcal{G}_j'(x)| \le \frac{5b}{4}x^{-3/8} + \frac{5}{8}x^{-13/8}$ for $j = 1, 2$ for $x \in D_1$, (21), (12) imply:

$$\left|w'(x) - w_0'(x)\right| \le \frac{1}{a}\sum_{j=1}^{2}|\mathcal{G}_j'(x)|\int_x^\infty |t\mathcal{G}_{3-j}|\left|N(w,t) - N_0(t)\right|dt \le 0.29x^{-11}. \quad \square$$

Recall y_0 is defined as:

$$y_0(x) =: \sqrt{\frac{x}{6}}\left[1 + \frac{1}{8\sqrt{6}}x^{-5/2} - \frac{49}{768}x^{-5} + \frac{1225}{1536\sqrt{6}}x^{-15/2} + w_0(x)\right]. \tag{28}$$

On using Lemma 3.4, (28) and (17) leads to the main result of this section:

Corollary 3.5. *The tritronquée solution has the representation $y = y_0 + E$ where $\left|E(x)\right| \le (0.00682)\|w_0\|_{10}x^{-43/4}$ and $\left|E'(x)\right| \le (0.126)x^{-21/2}$ on D_1. In particular, $\left|E(L^+)\right| \le 5.625 \times 10^{-10}$ and $|E'(L^+)| \le (2.12) \times 10^{-9}$.*

Proof. By construction, w satisfying (20) also satisfies the differential equation (18) with asymptotic condition $w(x) = o(x^{-5/8})$ as $x \to \infty$. On substituting $y = y_0 + \sqrt{\frac{x}{6}}(w - w_0)$, it follows y satisfies (1) with the asymptotic condition (2). Since such a solution is unique, $y = y_0 + \sqrt{\frac{x}{6}}(w - w_0)$ must be the tritronquée solution in D_1. The bounds on the error $E = y(x) - y_0(x) = \sqrt{\frac{x}{6}}(w - w_0)$ and its derivative follow from those satisfied by $w - w_0$ and its derivative, and the bounds on $\|w_0\|_{10}$. □

4. Analysis on $D_2 = [L_0, L)$

Remark 4.1. Since a double pole singularity of y at x_p close to x_0 is expected, it can potentially cause accuracy problems even on domain D_2. Hence, it is better to introduce a function u with less variation than y in D_2.

We define

$$E(x) = y(x) - y_0(x). \tag{29}$$

Substitution of (29) into (1) yields

$$\mathcal{L}E := E'' + 12y_0E = -6E^2 - R, \tag{30}$$

where R is the remainder $R(x) := y_0''(x) + 6y_0^2(x) - x$. Further, imposing C^1-transition of $y_0 + E$ at $x = L$ from left and right implies

$$E(L^-) = E(L^+) + y_0(L^+) - y_0(L^-) \quad , \quad E'(L^-) = E'(L^+) + y_0'(L^+) - y_0'(L^-). \tag{31}$$

Remark 4.2. The explicit form of y_0 enables exact evaluation of y_0 and all its derivatives, as well as R and its integral.

Definition 4.3. We define

$$\mathcal{R}(x) = \int_L^x [-R]dt \tag{32}$$

Lemma 4.4. *In the domain $\mathcal{D}_2$,*

$$\|\mathcal{R}\|_\infty \le 3.75 \times 10^{-9} \tag{33}$$

$$y_0(x) > 0 \quad , \quad y_0'(x) > 0 \tag{34}$$

Proof. We use a straightforward inequality (as stated in Lemma A.2 in the appendix) by partitioning the domain D_2 into n equal segments $\left\{\left[x_j, x_{j+1}\right]\right\}_{j=0}^{n-1}$. We note that both the integrals $\int_{x_{j-1}}^{x_j} \left[\mathcal{R}'(x)\right]^2 dx$ and point values $\mathcal{R}(x_j)$ can be determined explicitly. Thus checking bounds

on $\|\mathcal{R}\|_\infty$ is easily facilitated resulting in (33) by choosing[5] $n = 20$. Using Lemma A.1, we have for $x \in [x_j, x_{j+1}]$, $y_0(x) > \frac{1}{2}\left[y_0(x_j) + y_0(x_{j+1})\right] - \frac{1}{2}\sqrt{x_j - x_{j-1}}\|y_0'\|_{L^2[x_{j-1},x_j]}$ and since the point values of y_0 as well as explicit L_2 integral can be explicitly determined, we can check condition $y_0(x) > 0$ in each subinterval. The same procedure was repeated for y_0' to obtain (34). □

From (30), the two independent solutions are denoted as G_1, G_2 satisfying

$$G_j'' + 12y_0G_j = 0, \quad \text{for} \quad j = 1, 2 \tag{35}$$

with initial conditions $G_1(L^-) = 1$, $G_1'(L^-) = 0$, $G_2(L^-) = 0$ and $G_2'(L^-) = 1$.

Lemma 4.5. $\|G_1'\|_\infty \le 3.391$, $\|G_1\|_\infty \le 3.775$, $\|G_2'\|_\infty \le 1$ *and* $\|G_2\|_\infty \le 1.114$ *on* D_2.

Proof. On multiplication by $2G_j'$, integration from L to x and using integration by parts, (35) gives

$$G_j'^{\,2}(x) + 12y_0(x)G_j^2(x) + 12\int_x^L y_0'(t)G_j^2(t)dt = \mathcal{G}_j'^{\,2}(L) + 12y_0(L)\mathcal{G}_j^2(L) \tag{36}$$

Using $y_0, y_0' > 0$, and initial conditions on $\mathcal{G}_j$, (36) implies

$$G_1'(x)^2 + 12y_0(x)G_1(x)^2 \le 12y_0(L), \tag{37}$$

$$G_2'(x)^2 + 12y_0(x)G_2(x)^2 \le 1, \tag{38}$$

$|G_1'| \le \sqrt{12y_0(L)} \le 3.391$ and $|G_2'| \le 1$ are immediate. To find bounds on G_1, G_2, it is convenient to partition D_2 into two intervals $[L_0, \gamma_0)$ and $[\gamma_0, L)$, where γ_0 will be chosen appropriately. Using bounds on $|G_1'|$ in (37), implies

$$|G_1(x)| \le \frac{\sqrt{y_0(L)}}{\sqrt{y_0(x)}} \text{ when } \gamma_0 \le x \le L,$$

$$|G_1(x)| \le \int_x^{\gamma_0} |G_1'(x)|dx + |G_1(\gamma_0)| \le (\gamma_0 - x)\sqrt{12y_0(L)} + \frac{\sqrt{y_0(L)}}{\sqrt{y_0(\gamma_0)}} \text{ when } L_0 \le x \le \gamma_0$$

Since y_0 is monotonically increasing (see (34)), it follows from above that for any $x \in D_2$

$$\left|G_1(x)\right| \le (\gamma_0 - L_0)\sqrt{12y_0(L)} + \frac{\sqrt{y_0(L)}}{\sqrt{y_0(\gamma_0)}} \tag{39}$$

Similarly, using (38), we obtain for any $x \in D_2$,

[5] $n = 20$ gives a bound within about two percent of the graphically observed bound on $\mathcal{R}$; $n = 10$ gives only a 10 percent higher value.

$$\left|G_2(x)\right| \le \frac{1}{\sqrt{12 y_0(\gamma_0)}} + (\gamma_0 - L_0) \tag{40}$$

From explicit evaluation with $\gamma_0 = -\frac{16}{100}$, we get the bounds in the Lemma statement. □

Using variation of parameter, it follows from (30) that

$$\begin{aligned} E(x) &= \sum_{j=1}^{2} (-1)^{j+1} G_j(x) \int_L^x G_{3-j}(t)[-6E^2(t) - R(t)]dt + \alpha_1 G_1(x) + \beta_1 G_2(x) \\ &=: \mathcal{N}[E](x) \end{aligned} \tag{41}$$

where $\alpha_1 = E(L^+) + y_0(L^+) - y_0(L^-)$ and $\beta_1 = E'(L^+) + y_0'(L^+) - y_0'(L^-)$. Corollary 3.5 and small mismatch of y_0 and y_0' (see Remark 2.4) on two sides of L immediately implies

Corollary 4.6. $|\alpha_1| \le (5.63) \times 10^{-10}$ *and* $\beta_1 \le (2.13) \times 10^{-9}$.

Definition 4.7. Define

$$E_0(x) = \mathcal{N}[0](x) = \alpha_1 G_1(x) + \beta G_2(x) + \sum_{j=1}^{2} (-1)^j G_j(x) \int_L^x G_{3-j}(t) R(t) dt \tag{42}$$

Lemma 4.8. $\|E_0\|_\infty \le (1.745) \times 10^{-7}$ *and* $\|E_0'\|_\infty \le (1.605) \times 10^{-7}$ *on* D_2.

Proof. Integration by parts and use of boundary conditions at $x = L$ in (42) leads to

$$\sum_{j=1}^{2} (-1)^j G_j(x) \int_L^x G_{3-j}(t) R(t) dt = \sum_{j=1}^{2} (-1)^j G_j(x) \int_L^x G_{3-j}'(t) \mathcal{R}(t) dt \tag{43}$$

It follows

$$\begin{aligned} \left|E_0(x)\right| &\le \left(\|G_1'\|_\infty \|G_2\|_\infty \| + \|G_1\|_\infty \|G_2'\|_\infty\right)(L - L_0)\|\mathcal{R}\|_\infty + |\alpha_1|\|G_1\|_\infty + |\beta_1|\|G_2\|_\infty \\ &\le (1.745) \times 10^{-7}, \\ \left|E_0'(x)\right| &\le \|\mathcal{R}\|_\infty + 2\|G_1'\|_\infty \|G_2'\|_\infty \|\mathcal{R}\|_\infty (L - L_0) + |\alpha_1|.\|G_1'\|_\infty + |\beta_1|.\|G_2'\|_\infty \\ &\le 1.605 \times 10^{-7}, \end{aligned}$$

and taking supremum over $x \in D_2$ gives the result. □

The Banach space $\mathcal{S}$ is defined as

$$\mathcal{S} := \{f \in C(D_2) : \|f\|_\infty < \infty\}. \tag{44}$$

Let $f_0 \in \mathcal{S}$ and $r \geq 0$. The ball $\mathcal{B}(f_0, r) \subset \mathcal{S}$ centered at f_0 with radius r is defined as:

$$\mathcal{B}(f_0, r) := \{f \in \mathcal{S} : \|f - f_0\|_\infty \leq r\}. \tag{45}$$

Lemma 4.9. *Let $\delta = (5.5) \times 10^{-5}$. Then there exists a unique fixed point E of $\mathcal{N}$ in $\mathcal{B}(E_0, \delta\|E_0\|_\infty)$. Moreover, $\|E\|_\infty \leq (1.75) \times 10^{-7}$ and $\|E'\|_\infty \leq (1.61) \times 10^{-7}$.*

Proof. Let $\mathcal{E} \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. Using Lemmas 4.5 and 4.8; (41) implies

$$\left|\mathcal{N}[\mathcal{E}](x) - E_0(x)\right| \leq 2\|G_1\|_\infty\|G_2\|_\infty (L - L_0)\, 6(1+\delta)^2\|E_0\|_\infty^2 \leq \delta\|E_0\|_\infty$$

and taking supremum over $x \in D_2$ gives $\mathcal{N} : \mathcal{B} \to \mathcal{B}$. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. From Lemmas 4.8 and 4.5, using $\|\mathcal{E}_1 + \mathcal{E}_2\|_\infty \leq 2(1+\delta)\|E_0\|_\infty$, (41) implies:

$$\left|\mathcal{N}[\mathcal{E}_1](x) - \mathcal{N}[\mathcal{E}_2](x)\right| \leq 2\|G_1\|_\infty\|G_2\|_\infty (L - L_0)\, 12(1+\delta)\|E_0\|_\infty \leq 1.1 \times 10^{-4}\|\mathcal{E}_1 - \mathcal{E}_2\|_\infty$$

and taking supremum over $x \in D_2$ implies $\mathcal{N}$ is contractive. Banach fixed point theorem implies the existence and uniqueness of such E. $\|E\|_\infty \leq (1+\delta)\|E_0\|_\infty \leq (1.75) \times 10^{-7}$ is immediate. On using Lemmas 4.8 and 4.5, the derivative of (41) leads to

$$|E'(x)| \leq \|E_0'\|_\infty + \left(\|G_1'\|_\infty.\|G_2\|_\infty + \|G_2'\|_\infty.\|G_1\|_\infty\right)(L - L_0)6\|E\|_\infty^2 \leq 1.61 \times 10^{-7},$$

and taking supremum over $x \in D_2$ implies the result on $\|E'\|_\infty$. □

Corollary 4.10. *In D_2, the tritronquée has the representation $y = y_0 + E$ where, $\|E\|_\infty \leq (1.75) \times 10^{-7}$ and $\|E'\|_\infty \leq (1.61) \times 10^{-7}$ on D_2.*

Proof. The solution E in Lemma 4.9 which satisfies integral equation (41) also satisfies (30) because of smoothness of G_1, G_2. Therefore, it immediately follows that $y = y_0 + E$ satisfies P-1. Also the initial conditions (31) on E ensure that $y_0 + E$ and its derivative at $x = L^-$ match with the tritronquée at $x = L^+$, i.e. when L is approached from D_1. From uniqueness, y must be the tritronquée solution. Error bounds follow from Lemma 4.9. □

5. Analysis on $D_3 = [x_0 + r, L_0)$

With $y = y_0 + E$, where y_0 is given by (14), E satisfies

$$E'' - \frac{12}{(x - x_0)^2}E = -6E^2 - 12P_u E - R. \tag{46}$$

Imposing continuity of $y_0 + E$ and its derivative as $x = L_0$ is approached from the left and right implies

$$E(L_0^-) = E(L_0^+) \quad , \quad E'(L_0^-) = E'(L_0^+)\,, \tag{47}$$

since expression for y_0 is the same in domain D_3 and D_2. It is observed that

$$G_1(x) = (x - x_0)^4 \text{ and } G_2(x) = \frac{1}{(x - x_0)^3}, \tag{48}$$

are two independent solutions of the associated homogenous differential equation $G'' - \frac{12}{(x-x_0)^2} G = 0$. Using variation of parameter, (46) implies

$$\begin{aligned} E(x) &= \sum_{j=1}^{2} (-1)^j \frac{G_j(x)}{7} \int_{L_0}^{x} G_{3-j}[-6E^2 - 12P_u E - R]dt + \alpha_2 G_1(x) + \beta_2 G_2(x) \\ &:= \mathcal{N}[E](x), \end{aligned} \tag{49}$$

where in order to satisfy (47),

$$7\alpha_2 = G_2(L_0)E'(L_0^+) - G_2'(L_0)E(L_0^+) \tag{50}$$

$$7\beta_2 = -G_1(L_0)E'(L_0^+) + G_1'(L_0)E(L_0^+) \tag{51}$$

Using bounds in Corollary 4.10, equation (50) and (51) immediately gives

Corollary 5.1. $|\alpha_2| \le 9.22 \times 10^{-9} =: \alpha_{2,M}$ *and* $|\beta_2| \le (9.76) \times 10^{-7} =: \beta_{2,M}$.

Lemma 5.2. *In the domain* $D_2 \cup D_3$, $P_u > 0$.

Proof. We note $x \in D_2 \cup D_3$ corresponds to $\tau \in [-1, 1)$. We consider an approximate expression

$$P_a(\tau) = \frac{335867}{539062} + \frac{835179584688}{1968351794375}\,\tau - \frac{7294680}{73240997}\,\tau^2 + \frac{60789}{1703279}\,\tau^3 \tag{52}$$

It can be checked that

$$P_u(\tau) - P_a(\tau) = \sum_{k=0}^{22} d_k \tau^k \ , \text{ with } \left| P_u(\tau) - P_a(\tau) \right| \le \sum_{k=0}^{22} |d_k| \le 0.039 \tag{53}$$

On the other hand, it can be checked

$$P_a'(\tau) = \frac{182367}{1703279} \left[\left(\tau - \frac{40}{43} \right)^2 + \left(\frac{44}{25} \right)^2 \right] > 0 \tag{54}$$

and therefore P_a is an increasing function implying

$$P_a(\tau) > P_a(-1) > 0.063 \tag{55}$$

Using (53) and (55), we obtain

$$P_u(\tau) > 0.063 - 0.039 > 0 \quad \square \tag{56}$$

Definition 5.3. Let γ be a real number. We define weighted norm $\|.\|_\gamma$ on $C(D_3)$:

$$\|f\|_\gamma := \sup_{x\in D_3} \left|(x-x_0)^\gamma f(x)\right| \text{ for } f \in C(D_3). \tag{57}$$

We also define $E_{0,1}$, $E_{0,2}$ so that $E_0 = \mathcal{N}[0] = E_{0,1} + E_{0,2}$, with

$$E_{0,1}(x) = \alpha_2 G_1(x) + \beta_2 G_2(x) \tag{58}$$

$$E_{0,2}(x) = \frac{1}{7} G_1(x) \int_{L_0}^{x} G_2(t) R(t) dt - \frac{1}{7} G_2(x) \int_{L_0}^{x} G_1(t) R(t) dt \tag{59}$$

Remark 5.4. Since G_1, G_2 and R are explicit and involve only a finite number of terms in powers of $(x-x_0)$, $E_{0,2}$ and $E'_{0,2}$ can be computed explicitly for any $x \in \mathbb{D}_3$.

Lemma 5.5. *For $\gamma = 3.2$, $\|E_0\|_\gamma \le 2.04 \times 10^{-6}$, $\|E'_0\|_\infty \le 1.23 \times 10^{-5}$.*

Proof. Using Corollary 5.1, and explicit representation of G_1, G'_1, G_2, G'_2 it is easy to prove that $(x-x_0)^\gamma \left(\alpha_{2,M}|G_1| + \beta_{2,M}|G_2|\right) = (x-x_0)^\gamma \left(\alpha_{2,M} G_1 + \beta_{2,M} G_2\right)$, attains its maximum value at $x = L_0$, while $\alpha_{2,M}|G'_1| + \beta_{2,M}|G'_2| = \alpha_{2,M} G'_1 - \beta_{2,M} G'_2$ attains its maximum at $x = x_0 + r$. Thus, it is readily checked $\|E_{0,1}\|_\gamma \le 2.03 \times 10^{-6}$, $\|E'_{0,1}\|_\infty \le 1.2245 \times 10^{-5}$. Applying Lemma A.2 to $(x-x_0)^\gamma E_{0,2}$, and $E'_{0,2}$ for subdivisions of the interval $[x_0 + r, L_0]$ with $n = 5$, we obtain $\|E_{0,2}\|_\gamma \le 2.3 \times 10^{-9}$, $\|E'_{0,2}\|_\infty \le 3.8 \times 10^{-8}$. Adding up, we get the bounds for $\|E_0\|_\gamma$, $\|E'_0\|_\infty$, given in the Lemma statement. □

Definition 5.6. Define Banach space

$$\mathcal{S} := \{f \in C(D_3) : \|f\|_\gamma < \infty\}. \tag{60}$$

For $f_0 \in \mathcal{S}$ and $\epsilon > 0$, define

$$\mathcal{B}(f_0, \epsilon) := \{f \in \mathcal{S} : \|f - f_0\|_\gamma \le \epsilon\}. \tag{61}$$

Lemma 5.7. *Let $\delta = 0.963$. Then for $\gamma = 3.2$, there exists a unique fixed point E of $\mathcal{N}$ in $\mathcal{B}(E_0, \delta\|E_0\|_\gamma)$. Moreover, $\|E\|_\gamma \le 4.01 \times 10^{-6}$ and $\|E'\|_\infty \le 3.76 \times 10^{-5}$ on D_3.*

Proof. Let $\mathcal{E} \in \mathcal{B}(E_0, \delta\|E_0\|_\gamma)$. Using the fact that $G_1, G_2, P_u > 0$ in D_3, (49) implies

$$\begin{aligned}\left|\left[\mathcal{N}[\mathcal{E}] - E_0\right](x-x_0)^\gamma\right| \le \|E\|_\gamma \sum_{j=1}^{2} \frac{(x-x_0)^\gamma G_j(x)}{7} \int_x^{L_0} G_{3-j}(t-x_0)^{-\gamma} [12 P_u] dt \\ + \|E\|_\gamma^2 \sum_{j=1}^{2} \frac{(x-x_0)^\gamma G_j(x)}{7} \int_x^{L_0} 6 G_{3-j}(t-x_0)^{-2\gamma} dt\end{aligned}$$

Since

$$Q(x) = \sum_{j=1}^{2} \frac{(x-x_0)^\gamma G_j(x)}{7} \int_x^{L_0} G_{3-j}(t-x_0)^{-\gamma}[12P_u]dt \ ,$$

$$T(x) = \sum_{j=1}^{2} \frac{(x-x_0)^\gamma G_j(x)}{7} \int_x^{L_0} 6G_{3-j}(t-x_0)^{-2\gamma} dt \ ,$$

and their derivatives can be explicitly calculated, and their upper bounds in D_2 determined by applying Lemma A.2 the interval $[x_0 + r, L_0]$ by partition it into five equal intervals ($n = 5$) results in the bounds $\left|Q(x)\right| \le 0.49$ and $\left|T(x)\right| \le 1$. Therefore, it follows that

$$\left|\big[\mathcal{N}[\mathcal{E}] - E_0\big](x-x_0)^\gamma\right| \le \Big[(1+\delta)(0.49) + (1+\delta)^2\|E_0\|_\gamma\Big]\|E_0\|_\gamma \le \delta\|E_0\|_\gamma \tag{62}$$

and taking supremum over $x \in D_3$ implies $\mathcal{N} : \mathcal{B} \to \mathcal{B}$. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathcal{B}(E_0, \delta\|E_0\|_\gamma)$. From similar arguments and $\|\mathcal{E}_1\|_\gamma + \|\mathcal{E}_2\|_\gamma \le 2(1+\delta)\|E_0\|_\gamma$, (49) yields:

$$\begin{aligned}\left|\big[\mathcal{N}[\mathcal{E}_1] - \mathcal{N}[\mathcal{E}_2]\big](x-x_0)^\gamma\right| &\le \Big[(0.49) + (12.2)(1+\delta)\|E_0\|_\gamma\Big]\|\mathcal{E}_1 - \mathcal{E}_2\|_\gamma \\ &\le \frac{1}{2} \times \|\mathcal{E}_1 - \mathcal{E}_2\|_\gamma\end{aligned}$$

and taking supremum over $x \in D_3$ implies $\mathcal{N}$ is contractive in $\mathcal{B}$. Banach fixed point theorem implies the existence and uniqueness of solution to $E = \mathcal{N}[E]$ in ball $\mathcal{B}$. Further for such a solution,

$$\begin{aligned}\left|E'(x)\right| \le \|E_0'\|_\infty + \|E\|_\gamma &\left\{\sum_{j=1}^{2} \frac{\left|G_j'(x)\right|}{7} \int_x^{L_0} G_{3-j}(t-x_0)^{-\gamma}[12P_u]dt\right\} \\ + \|E\|_\gamma^2 &\left\{6\sum_{j=1}^{2} \frac{\left|G_j'(x)\right|}{7} \int_x^{L_0} G_{3-j}(t-x_0)^{-2\gamma} dt\right\}\end{aligned} \tag{63}$$

Using explicit computation of each of the two terms within $\{\cdot\}$ at $x = x_0 + r$ and the bound on $\|E_0'\|_\infty$ in Lemma 5.5, $\|E'\|_\infty \le \|E_0'\|_\infty + 6.3\|E\|_\gamma + 12.3\|E\|_\gamma^2 \le 3.76 \times 10^{-5}$. □

The main result of this section is consequence of Lemma 5.7:

Corollary 5.8. *In D_3, the tritronquée solution has the representation $y = y_0 + E$, where $\|E\|_\gamma \le 4.01 \times 10^{-6}$ and $\|E'\|_\infty \le 3.76 \times 10^{-5}$. In particular, $\left|E(x_0 + r)\right| \le 1.26 \times 10^{-5}$ and $\left|E'(x_0 + r)\right| \le 3.76 \times 10^{-5}$.*

Proof. It is clear from the smoothness of G_1, G_2 in D_3 that the solution E to (49) guaranteed by Lemma 5.7 also satisfies (46), implying $y = y_0 + E$ satisfies P-1. The continuity conditions (47)

imply that $y_0 + E$ matches the tritronquée at $x = x_0 + r$. Since the solution to the initial value problem is unique, $y = y_0 + E$ must be the tritronquée solution. The error bounds follow from Lemma 5.7. □

6. Analysis on $D_4 = \{x \in \mathbb{C} : |x - x_0| = r, x \neq x_0 + r\}$

It is useful to introduce $\zeta = x - x_0$. Then, from expression of y_0 in Theorem 2.3, it follows that

$$Y_0(\zeta) := y_0(x_0 + \zeta) = -\frac{1}{\zeta^2} + \zeta^2 P(\zeta) \tag{64}$$

where $P(\zeta)$ given by (8). E is defined as:

$$E(\zeta) = y(x_0 + \zeta) - y_0(x_0 + \zeta) \tag{65}$$

Substitution of (65) and (64) into (1) yields:

$$E''(\zeta) + 12Y_0(\zeta)E(\zeta) = -6E(\zeta)^2 - R(\zeta), \tag{66}$$

where $R(\zeta)$ is the residual $R(\zeta) := Y_0''(\zeta) + 6Y_0(\zeta)^2 - (x_0 + \zeta)$. Requiring that as $\nu \to 0^+$ on $\zeta = re^{i\nu}$ continuity of $E(\zeta) + Y_0(\zeta)$ with solution found in D_3, we obtain conditions

$$E\left(re^{i0^+}\right) = E\left([x_0 + r]^+\right) + y_0\left([x_0 + r]^+\right) - Y_0\left(re^{i0^+}\right) \tag{67}$$

$$E'\left(re^{i0^+}\right) = E'\left([x_0 + r]^+\right) + y_0'\left([x_0 + r]^+\right) - Y_0'\left(re^{i0^+}\right) \quad . \tag{68}$$

Lemma 6.1. *In D_4, $\|R\|_\infty \leq 1.311 \times 10^{-6}$.*

Proof. Recognizing that $Y_0(\zeta)$ is a truncation of an exact series representation for a solution to P-1 upto a power of ζ^{19}, one obtains

$$R(\zeta) = \sum_{j=18}^{38} R_j \zeta^j \ , \tag{69}$$

implying from calculation of R_j that on $|\zeta| = r$,

$$\|R\|_\infty \leq \sum_{j=18}^{38} |R_j| r^j \leq 1.311 \times 10^{-6} \quad \square \tag{70}$$

Remark 6.2. Now we seek to find bounds on two independent solutions of the homogenous system $E'' + 12Y_0E = 0$ associated with (66). For this purpose, it is useful to note the geometric bound

$$\left|a_j\right| \leq \frac{1}{2^j} \tag{71}$$

for $0 \leq j \leq 17$ that can be verified through calculation, using recurrence relation (9).

Definition 6.3. Define

$$G_1(\zeta) = \sum_{n=0}^{\infty} A_n \zeta^{4+n} \ , \ G_2(\zeta) = \sum_{n=0}^{\infty} B_n \zeta^{n-3}, \tag{72}$$

where

$$A_0 = 1 \ , \ A_1 = A_2 = A_3 = 0 \ , \ A_n = -\frac{12}{n(n+7)} \sum_{k=0}^{\min\{n-4,17\}} a_k A_{n-4-k} \quad , \quad \text{for } n \geq 4 \tag{73}$$

$B_0 = 1 \ , \ B_1 = B_2 = B_3 = B_7 = 0 \ , \ $ and

$$\text{for } n \geq 4, n \neq 7, \quad B_n = -\frac{12}{n(n-7)} \sum_{k=0}^{\min\{n-4,17\}} a_k B_{n-4-k} \tag{74}$$

Remark 6.4.

$$G_1(\zeta) = \zeta^4 + \sum_{n=0}^{\infty} A_{n+4} \zeta^{n+8} \quad , \quad G_2(\zeta) = \zeta^{-3} + \sum_{n=0}^{\infty} B_{n+4} \zeta^{n+1} \tag{75}$$

Lemma 6.5. *For any integer $n \geq 1$,*

$$\left|A_n\right| \leq c_A \left(\frac{3}{4}\right)^n \quad , \quad \left|B_n\right| \leq c_B \left(\frac{3}{4}\right)^n \ , \tag{76}$$

where $c_A = 0.21$ and $c_B = 0.85$.

Proof. We checked the inequalities for the first twenty two coefficients $\{A_n, B_n\}_{n=1}^{22}$ through explicit calculations involving (73) and (74). Assume the inequality holds for $n \leq n_0$ for some $n_0 \geq 22$. Then, using bounds on a_k, and noting that (73) and (74) no longer involves A_0, we obtain

$$\begin{aligned} \left|A_{n_0+1}\right| &\leq \frac{12}{(n_0+1)(n_0+8)} c_A \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k \left(\frac{3}{4}\right)^{n_0-3-k} \\ &\leq \frac{36 c_A}{(n_0+1)(n_0+8)} \left(\frac{4}{3}\right)^4 \left(\frac{3}{4}\right)^{n_0+1} \leq c_A \left(\frac{3}{4}\right)^{n_0+1} \end{aligned} \tag{77}$$

So, the inequality holds for $n_0 + 1$. By induction it holds for all n. The same induction proof works for B_n after using $\frac{36}{(n_0+1)(n_0-6)} \left(\frac{4}{3}\right)^4 \leq 1$ for $n_0 \geq 22$. □

Lemma 6.6. *In D_4,*

$$\|G_1\|_\infty \leq r^4 + \frac{81 c_A r^8}{256 - 192r} \leq 0.249 \ , \ \|G_2\|_\infty \leq \frac{1}{r^3} + \frac{81 c_B r}{256 - 192r} \leq 3.32 \tag{78}$$

$$\|G_1'\|_\infty \leq 4r^3 + \frac{81 c_A r^7 (32 - 21r)}{64(4 - 3r)^2} \leq 1.48 \ , \ \|G_2'\|_\infty \leq \frac{3}{r^4} + \frac{81 c_B}{16(4 - 3r)^2} \leq 13.7 \tag{79}$$

Proof. We use the bounds in (76) in the expression for G_1, G_2 and their first derivatives obtained from (75) and use triangular inequality and the identities $\sum_{n=0}^{\infty}|z|^n = \frac{1}{1-|z|}$, $\sum_{n=0}^{\infty} n|z|^{n-1} = \left(\frac{1}{1-|z|}\right)^2$ for $|z|<1$. □

Lemma 6.7. *G_1 and G_2 are two linearly independent solutions to the homogenous differential equation $G''+12Y_0G=0$ on D_4 with Wronskian $W := G_1G_2' - G_2G_1' = -7$.*

Proof. That G_1 and G_2 satisfy $G''+12Y_0G=0$ is clear from substituting each of the series in (75) for G and noting that the recurrence relation obtained is the one we have for A_n and B_n in (73) and (74), whose geometric decay rate assures convergence in D_4. Since the Wronskian W must be a constant, it is evident from the small ζ behavior that $G_1 = \zeta^4(1+o(1))$, $G_1' = 4\zeta^3(1+o(1))$, $G_2 = \zeta^{-3}(1+o(1))$, $G_2' = -3\zeta^{-4}(1+o(1))$, it follows that $W=-7+o(1)$; since it is independent of ζ, $W=-7$, implying also that the two solutions G_1, G_2 are independent. □

Applying standard variation of parameter argument, it is clear that (66) implies

$$E(\zeta) = \sum_{j=1}^{2}(-1)^{j+1}G_j(\zeta)\oint_{x_0+r}^{\zeta}\frac{G_{3-j}}{7}\left[-6E^2-R\right]dz + \alpha_3 G_1(\zeta) + \beta_3 G_2(\zeta) := \mathcal{N}[E](\zeta), \tag{80}$$

where the integration $\oint_{x_0}^{\zeta}$ is integral from x_0 to ζ over D_4 in the counterclockwise direction, α_3 and β_3 are chosen to satisfy (67), implying

$$\alpha_3 G_1(r) + \beta_3 G_2(r) = E\left([x_0+r]^+\right) + y_0\left([x_0+r]^+\right) - Y_0\left(e^{i0^+}\right), \tag{81}$$

$$\alpha_3 G_1'(r) + \beta_3 G_2'(r) = E'\left([x_0+r]^+\right) + y_0'\left([x_0+r]^+\right) - Y_0'\left(e^{i0^+}\right). \tag{82}$$

Remark 6.8. It is to be noted that in the integral reformulation (80) of the differential equation (66), we are using the initial conditions at $\zeta = r$, inherited from the analysis of the tritronquée solution in D_3. *A priori*, we are not requiring that the solution E returns to itself when ν changes from 0 to 2π on the circle $\zeta = re^{i\nu}$. However, it is well-known that any P-1 solution is single valued, and therefore so must be the solution $y = y_0 + E$ to P-1. Since y_0 is manifestly single valued, so must be E.

Lemma 6.9. $|\alpha_3| \le 3.82\times 10^{-5}$ *and* $|\beta_3| \le 3.76\times 10^{-6}$.

Proof. We use (81) and (82) to solve for α_3 and β_3. From bounds on $E^+(x_0+r)$ and $E'^+(x_0+r)$ in Corollary 5.8, using small mismatch for y_0, y_0' at x_0+r (see Remark 2.4) and the bounds on G_1, G_1', G_2, G_2' from Lemma 6.7, we obtain the bounds α_3 ad β_3. □

Lemma 6.10. *Let E_0 denote $E_0 = \mathcal{N}[0]$. Then, $\|E_0\|_\infty \le 2.34\times 10^{-5}$ and $\|E_0'\|_\infty \le 1.15\times 10^{-4}$ on D_4.*

Proof. Using (80) we have

$$\left|E_0\right| \le \frac{4\pi r}{7}\|G_1\|_\infty\|G_2\|_\infty\|R\|_\infty + |\alpha_3|\|G_1\|_\infty + |\beta_3|\|G_2\|_\infty \le 2.34\times 10^{-5} \tag{83}$$

$$\begin{aligned}\left|E_0'\right| &\le \frac{2\pi r}{7}\left(\|G_1\|_\infty\|G_2'\|_\infty + \|G_2\|_\infty\|G_1'\|_\infty\right)\|R\|_\infty + |\alpha_3|\|G_1'\|_\infty + |\beta_3|\|G_2'\|_\infty \\ &\le 1.15\times 10^{-4} \quad \square\end{aligned} \tag{84}$$

Definition 6.11. Define Banach space

$$\mathcal{S} := \{f \in C(D_4) : \|f\|_\infty < \infty\}. \tag{85}$$

For $f_0 \in \mathcal{S}$ and $\epsilon > 0$, define

$$\mathcal{B}(f_0, \epsilon) := \{f \in \mathcal{S} : \|f - f_0\| \le \epsilon\}. \tag{86}$$

Lemma 6.12. *Let $\delta = 2\times 10^{-4}$. There exists a unique fixed point E of $\mathcal{N}$ in $\mathcal{B}(E_0, \delta\|E_0\|_\infty)$. Moreover, $\|E\|_\infty \le 2.35\times 10^{-5}$ and $\|E'\|_\infty \le 1.16\times 10^{-4}$ on D_4.*

Proof. Let $\mathcal{E} \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$.

$$\left|\mathcal{N}[\mathcal{E}](\zeta) - E_0(\zeta)\right| \le \frac{24}{7}\pi r\|G_1\|_\infty\|G_2\|_\infty(1+\delta)^2\|E_0\|_\infty^2 \le \delta\|E_0\|,$$

and taking supremum over $\zeta \in D_4$ implies $\mathcal{N} : \mathcal{B} \to \mathcal{B}$. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. From similar arguments and $\|\mathcal{E}_1\|_\infty + \|\mathcal{E}_2\|_\infty \le 2(1+\delta)\|E_0\|_\infty$, (80) yields:

$$\begin{aligned}\|\mathcal{N}[\mathcal{E}_1] - \mathcal{N}[\mathcal{E}_2]\|_\infty &\le \frac{48}{7}\pi r\|G_1\|_\infty\|G_2\|_\infty(1+\delta)\|E_0\|_\infty\|\mathcal{E}_1 - \mathcal{E}_2\|_\infty \\ &\le 3\times 10^{-4}\|\mathcal{E}_1 - \mathcal{E}_2\|_\infty\,,\end{aligned}$$

and therefore, $\mathcal{N}$ is contractive. Banach fixed point theorem implies the existence and uniqueness of such $E \in \mathcal{B}(E_0, \delta\|E_0\|_\infty)$. The following are immediate:

$$\begin{aligned}&\|E\|_\infty \le (1+\delta)\|E_0\|_\infty \le 2.35\times 10^{-5}, \\ &\|E'\|_\infty \le \|E_0'\|_\infty + \frac{12\pi r}{7}\left(\|G_1'\|_\infty\|G_2\|_\infty + \|G_2'\|_\infty\|G_1\|_\infty\right)(1+\delta)^2\|E_0\|_\infty^2 \\ &\qquad \le 1.16\times 10^{-4}. \quad \square\end{aligned}$$

Corollary 6.13. *The tritronquée solution to P-1 has the following representation in D_4, $y = y_0 + E$ where $\|E\|_\infty \le 2.35\times 10^{-5}$ and $\|E'\|_\infty \le 1.16\times 10^{-4}$ on D_4.*

Proof. Smoothness of G_1 and G_2 on domain D_4 implies that the solution E satisfying (80) also solves (66). It is directly checked that $y = y_0 + E$ solves P-1 and, because of (67), matches the tritronquée at the boundary point in D_3. Since the solution to P-1 with given initial condition is unique, $y = y_0 + E$ must be the tritronquée; bounds follow from Lemma 6.12. $\square$

7. The proof of the Theorem 2.3

Proof of Theorem 2.3. Corollaries 3.5, 4.10, 5.8 and 6.13 immediate imply the error bounds on $E = y - y_0$ for the tritronquée. Since it is well-known that regular solutions of P-1 must be analytic, it follows that y is also analytic in D. Also, any solution to P-1 is meromorphic, with locally convergent series representation (3) near each singularity x_p. Cauchy integral formula implies that the integral $-\frac{1}{2\pi i}\oint_{|\zeta|=r}\zeta y(x_0+\zeta)d\zeta$ equals to the number of singularities of $y(x)$ in $|x-x_0|<r$. From this observation and Corollary 6.13, we calculate

$$\left|1+\frac{1}{2\pi i}\oint\limits_{|\zeta|=r}\zeta y(x_0+\zeta)d\zeta\right| \le \left|1+\frac{1}{2\pi i}\oint\limits_{|\zeta|=r}\zeta y_0(x_0+\zeta)d\zeta\right| + r^2\|y-y_0\|_\infty \le 1.2\times 10^{-5} < 1, \tag{87}$$

implying that there is only one singularity x_p of the tritronquée y in the region $|x-x_0|<r$. Once again, Cauchy integral formula and (3) imply $-\frac{1}{4\pi i}\oint_{|\zeta|=r}\zeta^2 y(x_0+\zeta)d\zeta = x_p - x_0$, and since $\zeta^2 y_0(x_0+\zeta)$ is analytic in $|\zeta|\le r$, it follows that

$$|x_p - x_0| \le \left|-\frac{1}{4\pi i}\oint\limits_{|\zeta|=r}\zeta^2 E(\zeta)d\zeta\right| \le \frac{r^3}{2}\|E\|_\infty \le 4.1\times 10^{-6}. \tag{88}$$

Also, it is known [6] that the pole of the tritronquée solution closest to the origin is on the negative real axis. Since our analysis shows solution cannot blow up on D, it follows that the singularity x_p, whose location is estimated in (88), is on the negative real axis. □

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Appendix A

Lemma A.1. *Let $y \in C^1[a,b]$, then*

$$\|y - \frac{1}{2}(y(a)+y(b))\|_\infty \le \frac{1}{2}\sqrt{b-a}\|y'\|_{L_2(a,b)} \tag{89}$$

$$\|y\|_\infty \le \frac{1}{2}\left|y(a)+y(b)\right| + \frac{1}{2}\sqrt{b-a}\|y'\|_{L_2(a,b)} \tag{90}$$

Proof. Note that $y(x) = y(a) + \int_a^x y'(t)dt$ and $y(x) = y(b) - \int_x^b y'(t)dt$. Therefore,

$$y(x) = \frac{1}{2}(y(a)+y(b)) + \frac{1}{2}\int_a^x y'(t)dt - \frac{1}{2}\int_x^b y'(t)dt$$

Therefore, it follows that

$$\left| y(x) - \frac{1}{2}\left(y(a) + y(b)\right) \right| \leq \frac{1}{2}\int_a^b |y'(t)|$$

The bounds (89) immediately follow from using Cauchy–Schwartz inequality. The second part (90) follows from (89) simply from triangular inequality. □

Lemma A.2. *Let $y \in C^1[a,b]$, then it follows that*

$$\|y\|_\infty \leq \frac{1}{2}\max_{0\leq k\leq n-1}\left\{\frac{1}{2}\left|y\left(a+\frac{k}{n}(b-a)\right)+y\left(1+\frac{(k+1)}{n}(b-a)\right)\right|\right.$$
$$\left. + \sqrt{\frac{b-a}{n}}\left\{\int_{a+\frac{k}{n}(b-a)}^{a+\frac{(k+1)}{n}(b-a)} y'^2(t)dt\right\}^{1/2}\right\} \tag{91}$$

Proof. This follows from using previous Lemma on each interval $\left(a + \frac{k}{n}(b-a), a + \frac{(k+1)}{n}(b-a)\right)$ and then taking the maximum over the n intervals. □

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